



Every order heard. Every loop closed.

The voice recorder for cardiac arrests. It hears the whole resuscitation team, logs every drug, shock and rhythm, runs the ACLS clock and speaks up when an order was never confirmed back.

THE PROBLEM

A code is run entirely by voice, and **orders get lost.**

26%

of 387 spoken orders in 89 filmed resuscitations were closed-loop (repeated back).

El-Shafy et al., J Surg Educ 2018

3.6×

faster completion of orders that **were** closed-loop.

Same study, HR 3.6 (95% CI 2.5–5.3)

≈ 0.27

agreement between hospital records and observers on epinephrine timing.

J Clin Med 2023, cardiac ICU arrests

Hands are on the chest, the airway and the syringes. Someone must stop patient care to write. Tap-based recorder apps still need free hands, and **nothing notices an order that nobody confirmed.**

Every order becomes a loop that must close.

1 • ORDERED

“Give one milligram of epinephrine.”

2 • READ BACK

“Epi one milligram, drawing up.”

3 • GIVEN

“Epi's in.”

SILENCE FOR 10 S → AMBER • 15 S → SPOKEN

Nobody confirmed the order.

DIFFERENT VALUE READ BACK → RED

Ordered 300 mg · read back 150 mg

“Epinephrine one milligram ordered. Not acknowledged.”

“Check dose. Ordered amiodarone three hundred milligrams, read back one hundred fifty milligrams.”

It also runs the ACLS clock out loud: 2-minute cycles, the epinephrine window, shocks and a shockable-rhythm check. It answers **“CodeLoop, last epi kab diya tha?”** from its own record.

THE CRASH-CART SCREEN • LIVE REPLAY THROUGH ASSEMBLYAI

REPLAY

Ears live

Voice ready

1.2s lag

00:59

CODE TIME

Mute 2 min

Speaks: timers + loops

Record

End code

CPR CYCLE

01:21

TO RHYTHM CHECK

CYCLES 0

EPINEPHRINE

Next window in 02:51

last dose

3 min

5 min

Last dose 00:08 ago

RHYTHM • SHOCKS • DRUGS

RHYTHM

V-fib

shockable · 00:43 ago

SHOCKS

1

last 200 J · 00:40 ago

Epinephrine 1 mg00:08 ago

OPEN ORDERS

1

Amiodarone 300 mg

00:03 open

ORDERED

"Amiodarone 300 milligrams" — C

READ BACK DIFFERS

"Amio 150 pushing" — C

GIVEN

—

Ordered 300 mg · read back 150 mg

Confirm 300 mg

Confirm 150 mg

COMPLETED

00:54

Epinephrine 1 mg

confirm value

went unacknowledged

closed loop

00:22

Shock 200 J

no order heard

TAP TO LOG

Pause CPR

Shock 200 J

Epi 1 mg given

Amio 300 mg given

Rhythm VF

Rhythm PEA

Rhythm asystole

HEARD IN THE ROOM

Arjun: compressor

B

Resume compressions. 00:24

CPR RESUME

meds

Give 1 milligram of epinephrine. 00:31

ORDER epinephrine 1 mg

B

Is IV access in yet? 00:33

meds

Line is in the left arm, 20 gauge. 00:36

C

Sorry, epi, one palayshi release of mercury ki ririki ki postmenentum fil of fil fil philadelphia chinu kinne barmanian. Epi one milligram is in. 00:54

READ BACK epinephrine 1 mg ?

GIVEN epinephrine 1 mg ?

low confidence

C

Amiodarone 300 milligrams. 01:00

ORDER amiodarone 300 mg

C

Amio 150 pushing. 01:01

READ BACK amiodarone 150 mg

CodeLoop

Check dose. Ordered amiodarone three hundred milligrams, read back one hundred fifty milligrams.

Waiting for a gap in the room...

Ask CodeLoop: last epi? what's due?

Ask

Real screenshot: the amiodarone read-back conflict caught at 00:59, with CodeLoop's spoken line on the right

4

The debrief every mock code should have, in minutes.

- **Quality vs AHA targets:** first shock, first epinephrine, read-back rate and time, estimated CPR fraction. A dashboard tracks them across codes.
- **Second listen:** AssemblyAI async Universal-3.5 Pro re-transcribes the whole recording and confirms each live event, or flags it for review. **13 of 13 confirmed**
- **Spoken debrief:** a hands-free Voice Agent conversation with tool calls. It quotes facts from the record, never computing them, and saves the team's lessons in their own words.

“The order for epinephrine one milligram went unacknowledged for more than ten seconds. What would help?”

 **Code Record**

Replay · demo_tour.ward

Spoken debrief

Print

Started 9/25/2026, 10:26:22 PM · code 0956c674c052 · ended · outcome rosc

DURATION

01:19

SHOCKS

1

DRUGS GIVEN

2

CLOSED-LOOP ORDERS

100%

3 of 3 read back

CONFLICTS CAUGHT

1

SPOKEN PROMPTS

3

Record intact. 79 audit entries, each hash-chained to the one before.
head 43a3ad5fdc3681b875033d306a98d9db76f0c6b5a29b127fd3f65fa5798f1c81

RESUSCITATION QUALITY VS TARGETS

Time to first shock	0:16	≤ 2:00 for VF/pVT	met
Time to first epinephrine	0:45	≤ 5:00 (non-shockable)	
Compression fraction (est.) · from spoken pause/resume calls	85%	≥ 80%	met
Longest pause in compressions (est.) · 1 pause	9.6 s	< 10 s	met
Orders read back (closed loop)	3 of 3	every order	met
Median time to read-back	11 s	—	
Orders left unacknowledged ≥ 10 s	1	0	missed
Read-back conflicts caught · each one a potential dosing error caught before it was given	1	—	

Targets: AHA adult ACLS and Get With The Guidelines-Resuscitation. CPR timing is estimated from spoken calls.

SECOND LISTEN

AssemblyAI async Universal-3.5 Pro · Medical Mode · the whole recording

13 of 13 live events confirmed by an independent transcription of the whole recording.

TEAM DEBRIEF

the team's words, from the spoken debrief

Action

The meds nurse should say every order back out loud.

Action

We should always repeat the dose.

Keep

Keep the fast shocks.

Change

Always read back every dose and assign one person to the IV.

Summary.

The team identified the importance of strict medication read-backs and clear task assignment to manage communication during high-stress moments.

AssemblyAI in four roles, each **load-bearing**.

CAPABILITY	WITHOUT IT	MEASURED
Universal-3.5 Pro streaming + tight turn detection	The record lags the code	event latency $\approx$ 1.2 s
Medical Mode + ACLS keyterms + context prompt	Drug names and doses misheard	96% vs 78% critical entities; the dose conflict was missed without them
Word-level speaker labels	Order and read-back look like one person	Turns split at word-level speaker changes
Word confidence	A misheard dose silently becomes fact	Below 0.6 $\rightarrow$ UNCONFIRMED , listed for sign-off
Hindi-English code-switching	Indian teams' real talk lost	Cross-script phonetic layer maps Devanagari back to English
Voice Agent API (semantic barge-in)	A bot talks over the team leader	$\sim$0.8 s to speak; stops mid-word when interrupted
Async U3.5 Pro + Voice Agent tools , after the code	Mishearings go unchecked; a debrief bot invents numbers	Second listen 13 / 13 ; debrief facts only via <code>get_code_facts</code>

Models understand speech. **Deterministic code decides.**

AssemblyAI · ears

U3.5 Pro Streaming

Room audio from a crash-cart tablet. Speaker labels, Medical Mode, ACLS keyterms, far-field Voice Focus, English + Hindi.

CodeLoop core

Grammar → resolver → ACLS engine

Every event quotes the exact words it came from. Loops, timers, flags and every spoken number come from unit-tested, replayable code. **The model never picks a dose.**

AssemblyAI · voice

Voice Agent API

Speaks fixed lines only, waits for a gap in the room, stops when a clinician talks over it. Never hears the room's chatter.

Low ASR confidence on a dose → UNCONFIRMED, never a fact

Its own voice is recognised and never logged as the team

A misheard “ROSC” only pauses timers; a human ends the code

Answers come from the record, composed deterministically

Never recommends treatment: it keeps time, the leader decides

Hash-chained audit log; the record is rebuilt from it

Held-out mock codes, committed before they were measured.

LIVE ASSEMBLYAI	PRECISION	RECALL	LOOPS
Held-out set 1, first run	0.97	0.71	5 / 8
Held-out set 2, first run	0.82	0.56	7 / 9
Both, after general fixes	0.96	0.88	16 / 17
VF demo, production config	0.97	0.91	3 / 3

- Each fresh set found real bugs, including “**one hundred and fifty joules**” → **two shocks**. Fixed with general rules plus a regression test.
- When unsure, CodeLoop **misses** rather than invents: precision stays high.
- Remaining misses are mostly recognition errors (“two hundred” heard as “100”), exactly what the read-back loop exists to catch.
- Limitation: synthetic TTS voices so far. Real-voice testing in a sim centre is milestone one.

Start where every hospital already practises: **mock codes.**

1 • Simulation

Sim centres and resuscitation committees: quality scores, a second listen and a spoken debrief for every mock code. No patient data, no clinical reliance.

2 • Documentation

Live codes, silent or timers-only. The recorder nurse reviews and signs; FHIR and GWTG-Resuscitation registry export.

3 • Real-time assist

Spoken loop nudges during live codes, after clinical validation and a SaMD regulatory pathway.

290,000+

adult in-hospital cardiac arrests a year in the US, each one run by voice.

Andersen et al., JAMA 2019

≈ \$37M/yr

6,100 US hospitals × an assumed \$500 per hospital per month. India and EMS come next.

AHA Fast Facts 2026 · pricing is our assumption

≈ \$2.66

AssemblyAI cost per 30-minute code. About 48 arrests per hospital a year cost ≈ \$130 to record.

List prices; $290,000 \div 6,100$ hospitals



Codes are loud, fast and hands-full.
CodeLoop is the recorder that **never looks away.**

Every order heard. **Every loop closed.**

`github.com/TusharTechs/codeloop · live demo: see submission`